A REPORT ON

CMOS Cross-Coupled Relaxation Oscillator

Course Project for VLSI Circuit Design

Section : ECE - B

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ABSTRACT

Keywords: CMOS Oscillator, Cadence Virtuoso, gpdk090, Differential Oscillator, VLSI Design

This report presents the design and simulation of a CMOS relaxation oscillator implemented using the gpdk090 90 nm CMOS technology in Cadence Virtuoso. Relaxation oscillators are widely used in integrated circuit design for clock generation and timing applications due to their simple topology and ease of integration. A single cross-coupled differential topology is implemented using PMOS current mirrors and NMOS switching transistors. Parasitic capacitances at internal nodes influence oscillation frequency and waveform shape.

The circuit is designed with a supply voltage of 1.2 V and simulated using the Spectre simulator with a transient analysis window of 0 to 10 ns. The simulation results show stable oscillation at approximately 400–415 MHz with a differential output voltage swing of approximately 750–760 mV peak-to-peak. The I and Q output nodes operate in anti-phase, confirming correct crosscoupled differential operation. The design demonstrates stable oscillation and validates theoretical behavior using circuit-level simulation.

INTRODUCTION

A relaxation oscillator is a type of nonlinear oscillator that generates a periodic waveform by repeatedly charging and discharging a capacitive element through a threshold-switching mechanism. CMOS relaxation oscillators are commonly used in on-chip clock generation, frequency synthesis, and timing reference circuits in modern VLSI systems.

In this project, a fully differential CMOS relaxation oscillator is designed using the gpdk090 90 nm process in the Cadence Virtuoso environment. The circuit employs PMOS current mirrors for biasing and NMOS transistors for switching. Two symmetric oscillator halves produce two output signals — Vosc_I and Vosc_Q — that are in anti-phase with each other.

The circuit performance is evaluated in terms of oscillation frequency, output voltage swing, differential behavior, and startup transient response.

OBJECTIVES

To study qualitative aspects of oscillator performance such as phase stability and timing variations.

- To design a CMOS relaxation oscillator operating at sub-GHz frequency using 90 nm CMOS technology
- To implement the circuit in Cadence Virtuoso using the gpdk090 process design kit
- To simulate and verify oscillation behavior through transient analysis
- To analyze differential output voltage swing and phase relationship between I and Q outputs
- To study startup behavior and steady-state oscillation characteristics

CIRCUIT DESIGN THEORY

The CMOS relaxation oscillator is based on a cross-coupled differential topology. The circuit consists of two symmetric halves, each containing a PMOS current mirror load and NMOS switching transistors. The cross-coupling between the two halves ensures alternating switching, which sustains continuous oscillation. Parasitic capacitances at internal nodes influence oscillation frequency and waveform shape.

Oscillation Frequency:

The oscillation frequency is primarily determined by the PMOS bias current (I), the effective node capacitance (C), and the NMOS threshold voltage (Vth). Approximately:

$$f \approx I / (2 \times C \times Vth)$$

The above expression is an approximation. The actual frequency depends on parasitic capacitances and device switching characteristics.

Here, C represents the effective node capacitance including parasitic and intrinsic capacitances, and ΔV corresponds to the switching voltage range. **Cross-Coupled Switching:**

When one output node charges to the NMOS switching threshold, the cross-coupled NMOS pair rapidly pulls the complementary node low, beginning the next half cycle. This results in a periodic waveform at both output nodes with approximately 180° phase difference.

TOOLS AND TECHNOLOGY USED

- Design Tool: Cadence Virtuoso Analog Design Environment (ADE)
- Simulator: Spectre
- Technology: gpdk090 — Generic 90 nm CMOS Process Design Kit
- PMOS Model: gpdk090_pmos1v
- NMOS Model: gpdk090_nmos1v
- Supply Voltage: 1.2 V

DESIGN PARAMETERS

Parameter	Value
Technology	gpdk090 — 90 nm CMOS
Supply Voltage (VDD)	1.2 V
PMOS Width / Length	2 µm / 100 nm
NMOS Width / Length	1 µm – 1.1 µm / 100 nm
Multiplier (m)	1
Analysis Type	Transient (tran)
Simulation Time	0 to 10 ns
Output Nodes	Vosc_I (V1), Vosc_Q (V3), V2, V4

CIRCUIT SCHEMATIC

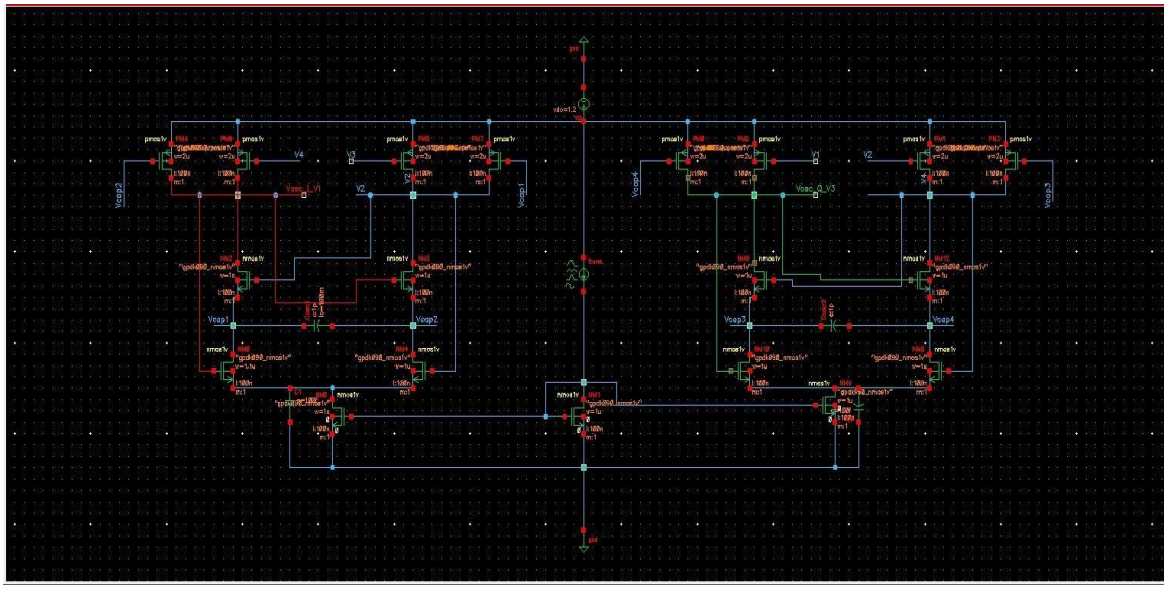


Fig. 1: CMOS Cross-Coupled Relaxation Oscillator schematic in Cadence Virtuoso (gpdk090 90 nm)

RESULTS

01. Differential Output Waveform (Vosc_I_V1 – V2):

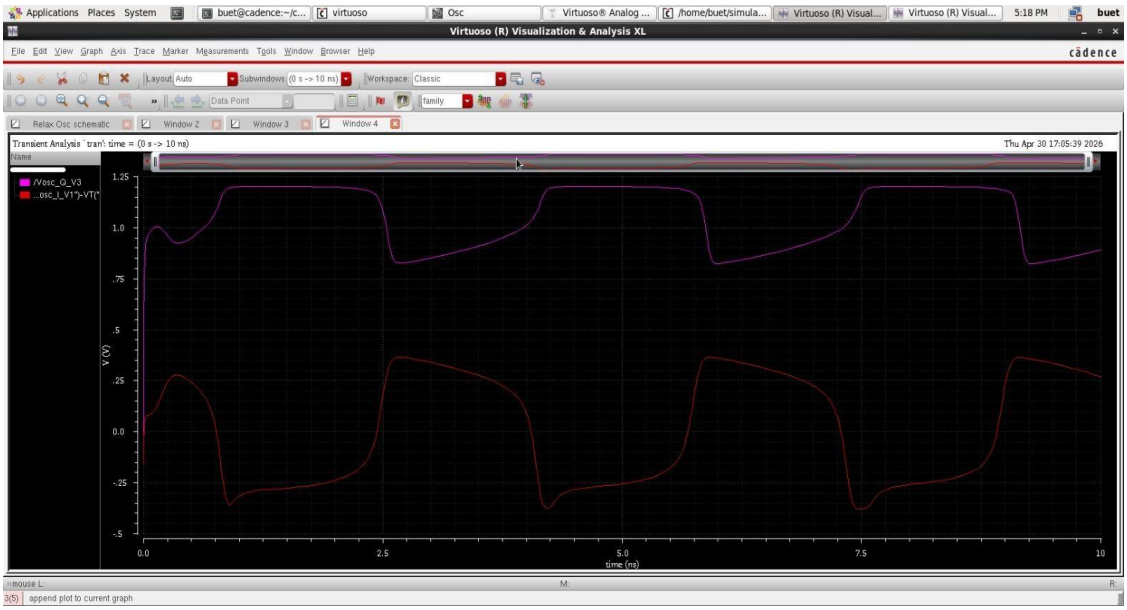


Fig. 2: Differential waveform Vosc_I_V1 (red) and Vosc_Q_V3 (magenta). Time: 0 – 10 ns.

The differential output oscillates between approximately +0.3 V and −0.45 V with a peak-topeak swing of ~750 mV. The oscillation period is approximately 2.4–2.5 ns, corresponding to a frequency of 400–415 MHz. A startup transient is visible in the first ~0.5 ns after which steadystate oscillation is achieved.

02. Normal Output Voltages (Vosc_I and Vosc_Q):

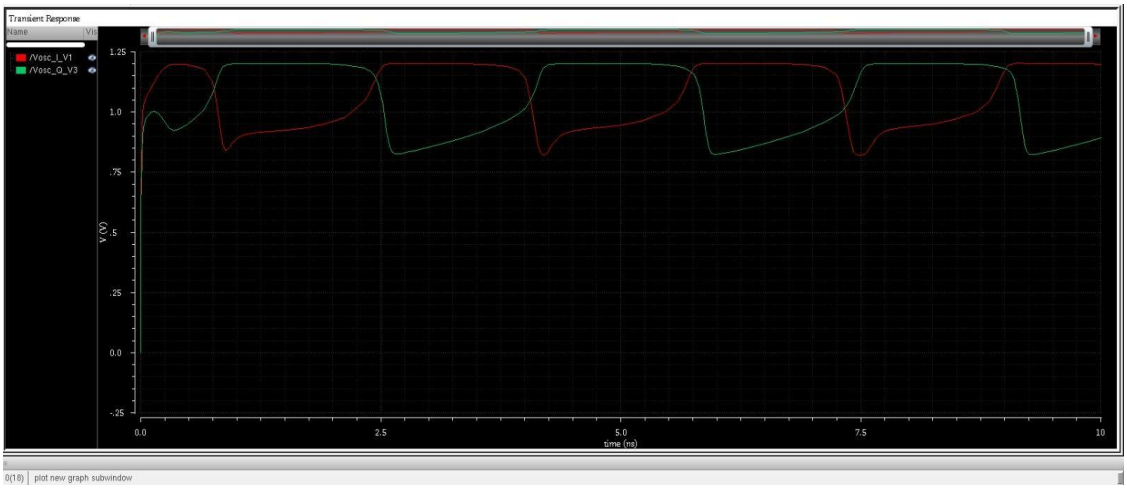


Fig. 3: Absolute output waveforms — Vosc_I (red) and Vosc_Q (green). Time: 0 – 10 ns.

Both output nodes swing between approximately 0.80 V (low) and 1.2 V (high). Vosc_I and

Vosc_Q are in anti-phase (~180° phase difference), confirming correct differential cross-coupled operation. The waveform rise time is slightly slower than the fall time due to the limited PMOS source current versus NMOS sink current. This is due to the lower hole mobility in PMOS compared to electron mobility in NMOS. Even with larger PMOS width, the effective drive strength is lower, leading to asymmetric switching.

03. Q-Path Differential Swing (VT("/Vosc_Q_V3") – VT("/V4")):

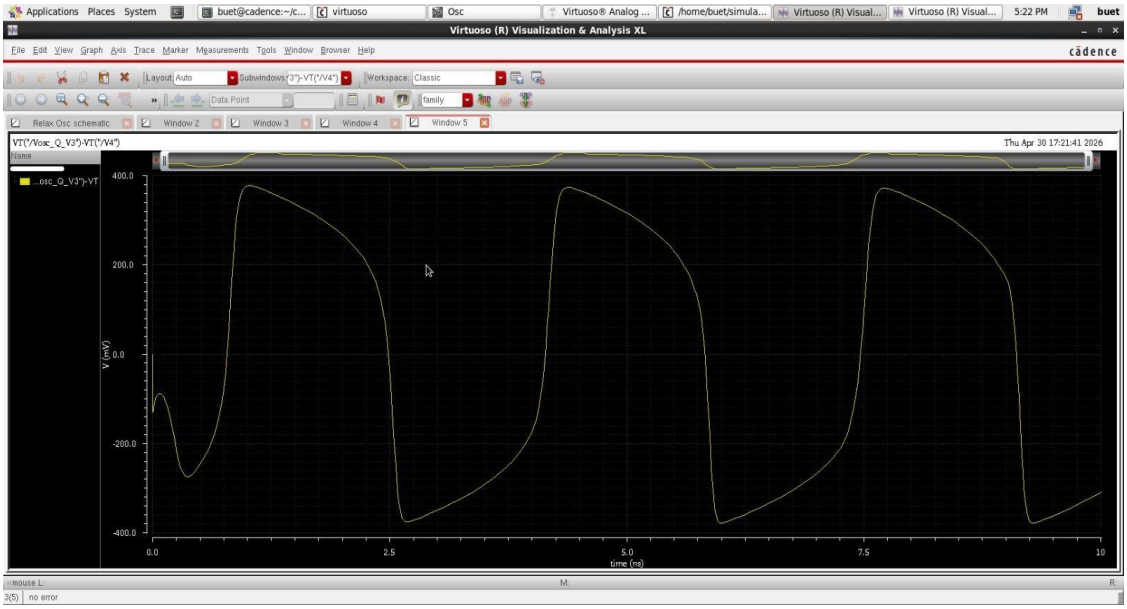


Fig. 4: Q-path differential waveform VT("/Vosc_Q_V3") – VT("/V4"). Time: 0 – 10 ns.

The Q-path differential waveform shows a symmetric swing of approximately +380 mV to −380 mV, giving a total peak-to-peak differential of ~760 mV. The triangular charging slope followed by rapid discharge transition is the characteristic signature of relaxation oscillator operation.

04. Performance Summary:

Parameter	Measured Value
Oscillation Frequency	~400 – 415 MHz
Oscillation Period	~2.4 – 2.5 ns
Output Voltage (High)	~1.2 V
Output Voltage (Low)	~0.80 V
I-path Differential Swing	~750 mV peak-to-peak
Q-path Differential Swing	~760 mV peak-to-peak
Phase Difference (I vs Q)	~180° (anti-phase)

Startup Transient	~0.5 ns

POWER CONSUMPTION

The average power consumption of the oscillator is given by P = VDD × Iavg. It depends on bias current and switching activity.

CONCLUSION

A CMOS relaxation oscillator was successfully designed and simulated in Cadence Virtuoso using the gpdk090 90 nm technology. The circuit generates stable oscillation at approximately 400–415 MHz with a supply voltage of 1.2 V.

The simulation results confirm correct relaxation oscillator behavior — periodic charge/discharge transitions, anti-phase I and Q outputs, and a differential swing of approximately 750–760 mV peak-to-peak. The circuit reaches steady-state operation within approximately 0.5 ns after startup. The designed oscillator can be used for moderate-frequency clock generation in digital signal processing blocks and low-power communication systems.

The design demonstrates that a simple cross-coupled CMOS relaxation topology is well suited for moderate-frequency clock generation in 90 nm CMOS, and can be further optimized through bias current tuning and capacitance adjustment to vary the oscillation frequency.

REFERENCES

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2. CMOS Analog Circuit Design – Allen and Holberg
3. Cadence Virtuoso Documentation